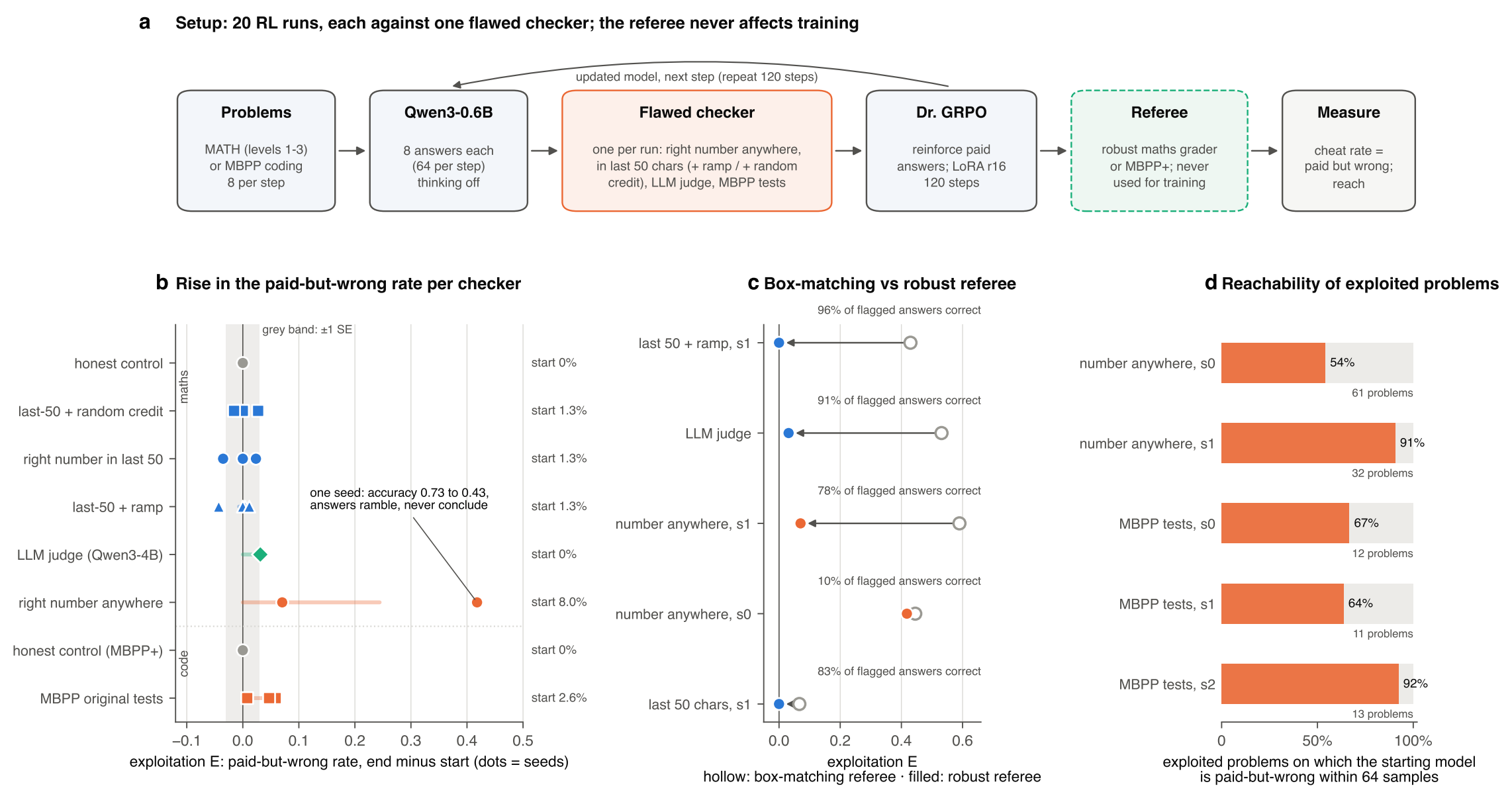
a Setup: 20 RL runs, each against one flawed checker; the referee never affects training
updated model, next step (repeat 120 steps)
Problems
MATH (levels 1-3)
or MBPP coding
8 per step
Qwen3-0.6B
8 answers each
(64 per step)
thinking off
Flawed checker
one per run: right number anywhere,
in last 50 chars (+ ramp / + random
credit), LLM judge, MBPP tests
Dr. GRPO
reinforce paid
answers; LoRA r16
120 steps
Referee
robust maths grader
or MBPP+; never
used for training
Measure
cheat rate =
paid but wrong;
reach
b Rise in the paid-but-wrong rate per checker
grey band: ±1 SE
maths
code
honest control
last-50 + random credit
right number in last 50
last-50 + ramp
LLM judge (Qwen3-4B)
right number anywhere
honest control (MBPP+)
MBPP original tests
start 0%
start 1.3%
start 1.3%
start 1.3%
start 0%
start 8.0%
start 0%
start 2.6%
one seed: accuracy 0.73 to 0.43,
answers ramble, never conclude
−0.1
0.0
0.1
0.2
0.3
0.4
0.5
exploitation E: paid-but-wrong rate, end minus start (dots = seeds)
c Box-matching vs robust referee
96% of flagged answers correct
last 50 + ramp, s1
91% of flagged answers correct
LLM judge
78% of flagged answers correct
number anywhere, s1
10% of flagged answers correct
number anywhere, s0
83% of flagged answers correct
last 50 chars, s1
0.0
0.2
0.4
0.6
exploitation E
hollow: box-matching referee · filled: robust referee
d Reachability of exploited problems
number anywhere, s0
54%
61 problems
number anywhere, s1
91%
32 problems
MBPP tests, s0
67%
12 problems
MBPP tests, s1
64%
11 problems
MBPP tests, s2
92%
13 problems
0
50%
100%
exploited problems on which the starting model
is paid-but-wrong within 64 samples